

Additional text 4: do the conclusions depend on how the runs are normalised?

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Question

GA_33 runs were median-normalised, which removed a rise of about 0.9 log2 units in run medians from 35 to 43 °C (a loading or global-binding difference that the data cannot resolve). The salt workbook was used as delivered. Do the conclusions survive other choices?

What can and cannot change

Median normalisation, normalisation to the bait's own signal, and no normalisation all subtract a single constant from each run. Across-protein comparisons are then unchanged, except for proteins missing from some runs, whose replicate means draw on different runs. These three choices are therefore a weak test, included for completeness. What could change the results is a difference in **dynamic range** between runs: if, say, the 43 °C runs compressed the intensity scale, abundant proteins would appear to fall and scarce ones to rise, which would mimic the strong dependence of the heat response on abundance. Two normalisations remove such differences: **scale** (each run centred and scaled to a common interquartile range) and **quantile** (each run mapped onto a common intensity distribution).

Results

1. Dynamic range. In GA_33 the 43 °C runs are slightly *wider*, not compressed (interquartile range 1.59 and 1.69 against 1.48 and 1.55 at 35 °C, for DNAJA1 and DNAJB11; standard deviation ratio 1.05). In the salt workbook the runs had already been median-aligned upstream (run medians 12.51-12.63), and the 75 and 150 mM runs are mildly narrower than the 0 mM runs (1.51 and 1.55 against 1.64).

2. GA_33 under five normalisations.

Norm.	Target	r vs median	Reliab.	rho, abund.	ESMC R2	Class effect
median	heat 43 °C	1	0.992	-0.43	25.3 %	TM -0.52
none		1.000	0.992	-0.43	25.3 %	TM -0.52
bait		1.000	0.991	-0.43	25.3 %	TM -0.52
scale		0.997	0.991	-0.50	24.4 %	TM -0.53
quantile		0.996	0.991	-0.50	24.1 %	TM -0.53
median	stau. 37 °C	1	0.755	-0.04	6.5 %	kinases -1.04 sd
scale		0.999	0.753	-0.11	6.3 %	kinases -1.03 sd
quantile		0.998	0.752	-0.09	6.3 %	kinases -1.03 sd
median	DNAJB11 pref.	1	0.959	+0.12	44.2 %	signal +0.35, nuc. -0.31
scale		0.993	0.959	+0.01	44.2 %	signal +0.36, nuc. -0.30
quantile		0.992	0.957	+0.00	43.5 %	signal +0.36, nuc. -0.30

ESMC R2: ridge, final layer for heat and preference, layer 50 for staurosporine, folds grouped by sequence cluster, one penalty (10^4) throughout. Heat 37 °C behaves like heat 43 °C (reliability 0.86-0.87, R2 12.0-12.5 %).

3. The salt titration under four normalisations.

Norm.	Target	r vs delivered	Reliab.	rho, abund.	ESMC R2	TM effect
as delivered	150 mM	1	0.992	-0.44	18.2 %	+0.46
median		1.000	0.992	-0.44	18.2 %	+0.46
scale		0.997	0.992	-0.36	18.3 %	+0.45
quantile		0.996	0.992	-0.36	18.1 %	+0.45
as delivered	75 mM	1	0.986	-0.41	16.0 %	+0.32
scale		0.988	0.986	-0.25	15.7 %	+0.31
quantile		0.987	0.986	-0.24	15.6 %	+0.31

Conclusions

- **Every sequence result is robust to normalisation.** The targets keep at least 97 % of their variance in common with the original ($r \geq 0.987$), reliabilities by at most 0.008, sequence R2 by at most 1.2 points, and the class effects (transmembrane, kinases, signal peptide, nuclear) by at most 0.02.
- **The heat response's dependence on abundance is not a dynamic-range artefact.** Equalising the runs' dynamic range makes it stronger (-0.43 to -0.50), not weaker. It reflects a real change in the composition of the pull-downs between 35 and 43 °C: the 43 °C and 35 °C profiles correlate at about 0.6 with nearly equal spread (standard deviation ratio 1.05), and in that situation the change between them is necessarily anti-correlated with the starting level.
- **Part of the salt response's dependence on abundance is technical.** The salt runs are mildly compressed relative to the 0 mM runs, and removing that weakens the dependence from -0.44 to -0.36 (150 mM) and from -0.41 to -0.24 (75 mM). This changes nothing in the sequence results, which the main report had already shown to be independent of abundance.
- **The global 0.9 log2 rise from 35 to 43 °C** remains unresolvable from these data: loading and a real increase in total binding look identical in a pull-down. It affects no across-protein result.

Critical evaluation

- **The constant-offset normalisations are not a real test** (explained above); the conclusions rest on the scale and quantile comparisons.
- **Intensity-dependent normalisation was not used.** Methods such as loess on intensity would remove any dependence of the response on abundance by construction, which would erase the very thing under question rather than test it. The dynamic-range comparison is the appropriate test.
- **One ridge penalty.** Fixing the penalty at the value earlier fits chose keeps the comparison across normalisations exact; R2 values are a few tenths of a point below fully tuned ones (e.g. 25.3 % against 25.7 % for heat).
- **The salt workbook's upstream processing is unknown.** Its runs were already median-aligned; whether other corrections were applied cannot be determined from the file.

Methods files

code/ad04_normalisation.py (GA_33) and code/ad04b_salt_normalisation.py (salt). Results: data/ga_data/ad04_results.json, data/pd_data/ad04b_results.json.